Filter banks localization

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1 EnVar formulation

1.1 Standard EnVar formulation

For an ensemble N backgrounds $\left\{\mathbf{x}^1, \dots, \mathbf{x}^N\right\}$, with $\mathbf{x}^k \in \mathbb{R}^n$, the standard EnVar formulation is given by:

$$\mathbf{B} = \mathbf{X}\mathbf{X}^\mathrm{T} \circ \mathbf{L} \tag{1}$$

where:

- $\mathbf{B} \in \mathbb{R}^{n \times n}$ is the resulting background error covariance.
- $\mathbf{X} \in \mathbb{R}^{n \times N}$ contains the normalized ensemble perturbations:

$$X_{ij} = \frac{1}{\sqrt{N-1}} \left(x_i^j - \langle x_i \rangle\right) \tag{2}$$

  with $\langle x_i \rangle$ denoting the ensemble mean:

$$\langle x_i \rangle = \frac{1}{N} \sum_{j=1}^N x_i^j \tag{3}$$

- $\mathbf{L} \in \mathbb{R}^{n \times n}$ is the localization matrix.
- $\circ$ denotes a Schur product between matrices (element-by-element).

The square-root expression of (1), transforming the control vector $\mathbf{v} \in \mathbb{R}^m$ into a control increment $\delta\mathbf{x} \in \mathbb{R}^n$, is given by:

$$\delta\mathbf{x} = \sum_{j=1}^N \mathbf{X}_{:j} \circ \left(\mathbf{U}\mathbf{v}_j\right) \tag{4}$$

where:

- $\mathbf{X}_{:j} \in \mathbb{R}^n$ is the j[th] column of $\mathbf{X}$
- $\mathbf{U} \in \mathbb{R}^{n \times m}$ is a square-root of the localization $\mathbf{L}$ such that $\mathbf{U}\mathbf{U}^{\mathrm{T}} = \mathbf{L}$.
- $\mathbf{v}_j \in \mathbb{R}^{m/N}$ is the j[th] subvector of $\mathbf{v}$.

1.2 Generalized EnVar formulation

The standard EnVar formulation can be generalized by introducing two linear transforms $\mathbf{F} \in \mathbb{R}^{n \times \widehat{n}}$ and $\mathbf{F}^\dagger \in \mathbb{R}^{\widehat{n} \times n}$:

$$\widehat{\mathbf{B}} = \mathbf{F} \left(\left(\mathbf{F}^\dagger \mathbf{X}\right) \left(\mathbf{F}^\dagger \mathbf{X}\right)^{\mathrm{T}} \circ \widehat{\mathbf{L}} \right) \mathbf{F}^{\mathrm{T}} \tag{5}$$

$$\delta \mathbf{x} = \mathbf{F} \sum_{j=1}^{N} \left(\mathbf{F}^\dagger \mathbf{X}_{:j} \right) \circ \left(\widehat{\mathbf{U}} \widehat{\mathbf{v}}_j \right) \tag{6}$$

where:

- $\widehat{\mathbf{L}} \in \mathbb{R}^{\widehat{n} \times \widehat{n}}$ is the localization matrix in the transformed space.
- $\widehat{\mathbf{U}} \in \mathbb{R}^{n \times \widehat{m}}$ is a square-root of the localization $\widehat{\mathbf{L}}$ such that $\widehat{\mathbf{U}}\widehat{\mathbf{U}}^{\mathrm{T}} = \widehat{\mathbf{L}}$.
- $\widehat{\mathbf{v}}_j \in \mathbb{R}^{\widehat{m}/N}$ is the j[th] subvector of $\widehat{\mathbf{v}} \in \mathbb{R}^{\widehat{m}}$.

Let's analyse the asymptotic values if the ensemble size grows to infinity:

- The localization matrices $\mathbf{L}$ and $\widehat{\mathbf{L}}$ should converge to $\mathbf{1}_n$ and $\mathbf{1}_{\widehat{n}}$, respectively, where $\mathbf{1}_n \in \mathbb{R}^{n \times n}$ is a matrix filled with 1.
- As a consequence:

$$\lim_{N \to \infty} \mathbf{B} = \mathbf{X}\mathbf{X}^{\mathrm{T}} \tag{7}$$

$$\lim_{N \to \infty} \widehat{\mathbf{B}} = \left(\mathbf{F}\mathbf{F}^\dagger \mathbf{X}\right) \left(\mathbf{F}\mathbf{F}^\dagger \mathbf{X}\right)^{\mathrm{T}} \tag{8}$$

- To ensure consistency, linear transforms $\mathbf{F}$ and $\mathbf{F}^\dagger$ should not have an impact on the asymptotic background error covariance:

$$\lim_{N \to \infty} \mathbf{B} = \lim_{N \to \infty} \widehat{\mathbf{B}} \tag{9}$$

  so

$$\mathbf{F}\mathbf{F}^\dagger \mathbf{X} = \mathbf{X} \tag{10}$$

  Thus, $\mathbf{F}^\dagger$ should be a right-inverse of $\mathbf{F}$ in the ensemble perturbations space.

The content of $\mathbf{F}$ and $\mathbf{F}^\dagger$ can be very diverse:

- A change of variables: for instance towards a better-suited humidity variable.

- A inverse balance operator as in Clayton *et al.* (2012), where the localization is applied on unbalanced variables and the balance operator is finally applied to get the increment.

- A change of resolution as in Kleist and Ide (2014), where the ensemble resolution is lower than the final increment one. In this case, $\mathbf{F}$ is an interpolation from low to high resolution, and the low-resolution perturbations $\mathbf{X}_{\mathrm{LR}}$ can be thought as the application of a coarsening operator $\mathbf{F}^\dagger$ on virtual high-resolution perturbations $\mathbf{X}_{\mathrm{HR}}$ representing the same field:

$$\mathbf{F}\mathbf{X}_{\mathrm{LR}} = \mathbf{F}\mathbf{F}^\dagger\mathbf{X}_{\mathrm{HR}} = \mathbf{X}_{\mathrm{HR}} \tag{11}$$

2 Filter banks localization

2.1 The band-pass framework

We define two series of P linear transforms:

- the band-pass filters $\left\{\mathbf{H}_1, \dots, \mathbf{H}_P\right\}$, where $\mathbf{H}_p \in \mathbb{R}^{\widehat{n}_p \times n}$,

- the interpolators $\left\{\mathbf{S}_1, \dots, \mathbf{S}_P\right\}$, where $\mathbf{S}_p \in \mathbb{R}^{n \times \widehat{n}_p}$,

where $\widehat{n}_p$ is the space size of component p, and $\widehat{n} = \sum_{p=1}^{P} n_p$ is the total size.

Formally, the linear transform $\mathbf{F}^\dagger$ is defined as

$$\mathbf{F}^\dagger = \begin{pmatrix} \mathbf{H}_1 \\ \mathbf{H}_2 \\ \vdots \\ \mathbf{H}_P \end{pmatrix} \tag{12}$$

and its left-inverse as:

$$\mathbf{F} = \left(\mathbf{S}_1 \mathbf{S}_2 \dots \mathbf{S}_P\right) \tag{13}$$

For $1 \leq p \leq P-1$, the band-pass filters $\mathbf{H}_p$ are defined freely, but to ensure that $\mathbf{F}\mathbf{F}^\dagger = \mathbf{I}_n$, the last band-pass filter $\mathbf{H}_P$ is defined as the complement of the sum of all the previous band-pass filters interpolated at full resolution:

$$\mathbf{H}_P = \mathbf{I}_n - \sum_{p=1}^{P-1} \mathbf{S}_p \mathbf{H}_p \tag{14}$$

We also define the filtered components $\widehat{\mathbf{x}}_p \in \mathbb{R}^{\widehat{n}_p}$ as:

$$\widehat{\mathbf{x}}_p = \mathbf{H}_p \mathbf{x} \tag{15}$$

where $\mathbf{x}$ is the input vector.

2.2 Recursive construction

In this section, we build the band-pass filters $\mathbf{H}_p$ recursively from a series of $P-1$ filters $\left\{\mathbf{G}_1, \ldots, \mathbf{G}_{P-1}\right\}$, where $\mathbf{G}_p \in \mathbb{R}^{\widehat{n}_p \times n}$:

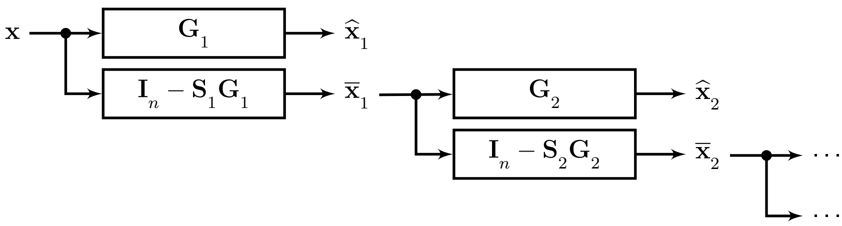
- The initial residual is given by the input state: $\overline{\mathbf{x}}_0 = \mathbf{x}$
- At each iteration $1 \leq p \leq P-1$:
  - The filter $\mathbf{G}_p$ is applied to the previous residual $\overline{\mathbf{x}}_{p-1} \in \mathbb{R}^n$ to produce the new filtered component $\widehat{\mathbf{x}}_p \in \mathbb{R}^{\widehat{n}_p}$:

$$\widehat{\mathbf{x}}_p = \mathbf{G}_p \overline{\mathbf{x}}_{p-1} \tag{16}$$

  - The new residual is computed from $\widehat{\mathbf{x}}_p$, interpolated at full resolution with $\mathbf{S}_p$:

$$\begin{aligned} \overline{\mathbf{x}}_p &= \overline{\mathbf{x}}_{p-1} - \mathbf{S}_p \widehat{\mathbf{x}}_p \qquad (17) \\ &= \left(\mathbf{I}_n - \mathbf{S}_p \mathbf{G}_p\right) \overline{\mathbf{x}}_{p-1} \qquad (18) \end{aligned}$$

This construction can be illustrated as follows:



Integrating recursively, the residual at iteration p becomes:

$$\begin{aligned} \overline{\mathbf{x}}_p &= \left(\mathbf{I}_n - \mathbf{S}_p \mathbf{G}_p\right)\left(\mathbf{I}_n - \mathbf{S}_{p-1} \mathbf{G}_{p-1}\right) \ldots \left(\mathbf{I}_n - \mathbf{S}_1 \mathbf{G}_1\right) \mathbf{x} \\ &= \prod_{q=1}^{p} \left(\mathbf{I}_n - \mathbf{S}_q \mathbf{G}_q\right) \mathbf{x} \qquad (19) \end{aligned}$$

and from (16), we obtain the filtered component at iteration p:

$$\widehat{\mathbf{x}}_p = \mathbf{G}_p \prod_{q=1}^{p-1} \left(\mathbf{I}_n - \mathbf{S}_q \mathbf{G}_q\right) \mathbf{x} \tag{20}$$

Combining equations (15) with (20), we get an equivalence between $\mathbf{H}_p$ and $\mathbf{G}_p$ for $1 \leq p \leq P-1$:

$$\mathbf{H}_p = \mathbf{G}_p \prod_{q=1}^{p-1} \left(\mathbf{I}_n - \mathbf{S}_q \mathbf{G}_q\right) \tag{21}$$

The last filtered component $\widehat{\mathbf{x}}_P$ is actually given by the last residual $\overline{\mathbf{x}}_{P-1}$

$$\widehat{\mathbf{x}}_P = \mathbf{H}_P \mathbf{x} = \overline{\mathbf{x}}_{P-1} \tag{22}$$

So equation (19) gives another expression of equation (14):

$$\mathbf{H}_P = \prod_{q=1}^{P-1} \left(\mathbf{I}_n - \mathbf{S}_q \mathbf{G}_q \right) \tag{23}$$

2.3 Low-pass and high-pass filters

The filters $\mathbf{G}_p$ can be:

- either low-pass filters, with $\widehat{n}_p$ increasing up to $\widehat{n}_P = n$,
- or high-pass filters, with $\widehat{n}_p$ decreasing from $\widehat{n}_1 = n$.

Often in practice, only low-pass filters are actually implemented, through a diffusion process or an equivalent smoother. However, the high-pass filter $\mathbf{G}_p^h$ can be built as the complement of the low-pass filter $\mathbf{G}_p^l$:

$$\mathbf{G}_p^h = \mathbf{I}_n - \mathbf{G}_p^l \tag{24}$$

But we should notice that in this case:

- All the components must have the same grid size $\widehat{n}_p = n$, so the size of the transformed space is $\widehat{n} = Pn$.
- It is not possible to benefit from interpolations: $\mathbf{S}_p = \mathbf{I}_n$

Figure 1 shows an example of spectra for explicit low-pass filters with Gaussian shapes, similar to the ones resulting from a diffusion process, and their complementary high-pass filters counterparts. The equivalent band-pass filters are also displayed in both cases. We can see that the band-pass filters have roughly the same structure in both cases, but are not identical.

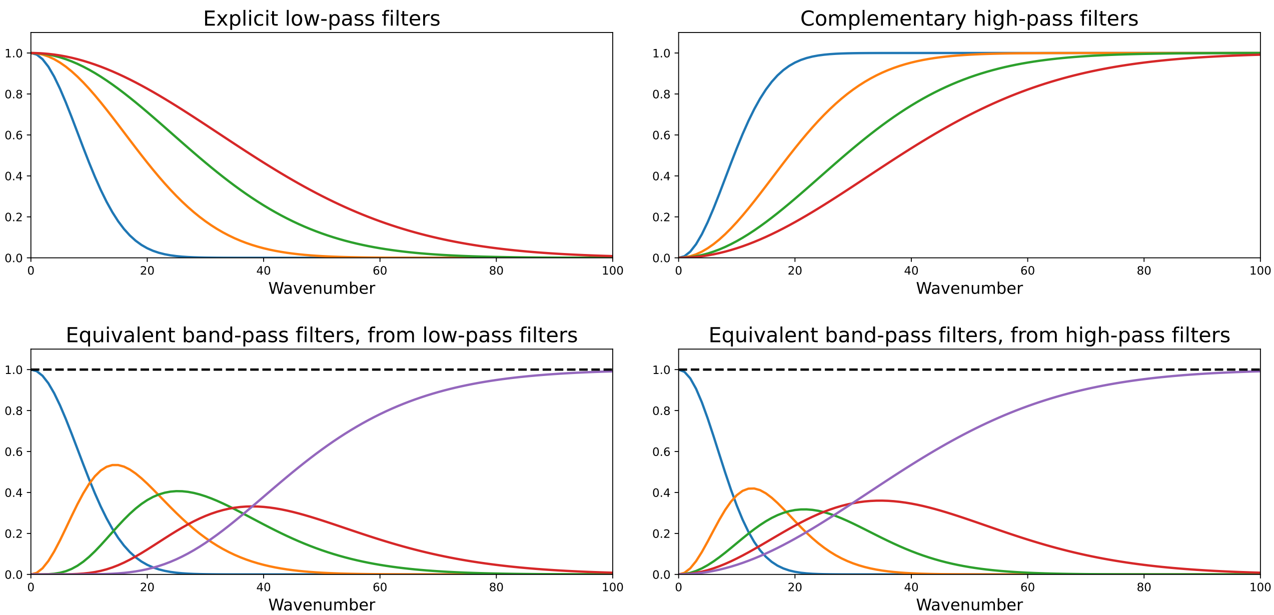


Figure 1: Spectra of explicit low-pass filters (top left) and their complementary high-pass filters counterparts (top right). The equivalent band-pass filters are displayed for both cases in the bottom panels, the dashed black line indicating the sum of all the band-pass filters.

2.4 Application

The standard scale-dependent localization (SDL) described in Buehner and Shlyaeva (2015) is a case where the band-pass filters are:

- either defined explicitely in spectral space, in a "wavelet-like" fashion,
- or built recursively from high-pass filters, themselves implemented as complements of diffusion-based low-pass filters.

To be continued...

References

Buehner M, Shlyaeva A. 2015. Scale-dependent background-error covariance localisation. *Tellus A* **67**(0).

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